Sumanth Veluvolu

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Professional Summary

Graduate student in Data Analytics with expertise in **RAG-powered AI systems**, large-scale data pipelines, **big data (90GB+)**, and cloud-native solutions. Delivering real-time insights and scalable **AI applications**, seeking to leverage **AI/ML** and data engineering expertise to build impactful, industry-scale solutions.

Professional Experience

Wintech Services Pvt. Ltd, Gen AI & Data Engineering Intern — Python, RAG, LangChain
Hyderabad, India May 2025 – Sep 2025

- Engineered **RAG-powered AI chatbots** by processing **90 GB+ of CSV data**, embedding into **FAISS vector databases, and integrating with LangChain (LLaMA 3)**, enabling real-time SKU-to-Model lookups and delivering query responses in < **1 second across 1M+ records with 99% uptime.**
- Developed a **scalable RAG application for 2,000+ CSV files**, delivering natural-language querying with fuzzy SKU matching, synonym recognition, and **advanced filtering (price, brand, category)**. Deployed via a Gradio chatbot interface, accelerating query resolution **from hours to seconds.**
- Analyzed **1,000+ product catalogs (millions of records)** with **PySpark and Pandas**, uncovering pricing trends, SKU performance metrics, and inventory turnover rates that informed **data-driven pricing and inventory strategies.**

Projects

AI-Powered Data Pipeline & RAG Chatbot — ***PySpark, Pandas, LangChain, FAISS, LLaMA3, Gradio***

- Reduced query resolution from **hours to seconds** by building a Gradio-based **real-time SKU lookup system**, enabling faster and more accurate decision-making.
- Achieved **sub-second retrieval across 1M+ SKUs with 99% uptime** by designing and deploying a large-scale RAG chatbot processing 90GB+ of data (3,000+ CSVs) with FAISS and LangChain (LLaMA3).

Big Data & ML for Smart Road Condition Analytics — ***PySpark, Random Forest, Gradient Boosting***

- Built a distributed **PySpark ML pipeline** capable of processing **1.2M+ pavement condition records 40% faster**, improving throughput from **8,000 to 11,500 records/minute**, ensuring scalability for city-scale road analytics.
- Achieved **92% prediction accuracy** across 20 pavement condition categories (18 predicted correctly on average), with AUC > 0.90, enabling **robust risk assessment for city-scale infrastructure planning.**

Restricting Unauthorized Access Using AWS Cloud — ***AWS Lambda, S3, API Gateway, IAM, CloudFront, DynamoDB***

- Engineered a secure, **serverless authentication system with geo-based access control**, dynamically blocking **500+ suspicious login attempts.**
- Achieved **99.9% system uptime** while strengthening **cloud security** posture and ensuring enterprise compliance.

Technical Skills

Languages and Databases: Python, SQL, C++, MongoDB, MySQL
Developer Tools: AWS, Databricks, RapidMiner, Git, Azure, Apache Spark
ML/AI Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, LangChain

Certifications

AWS Academy Data Engineering (2024) — AWS Academy Cloud Foundations (2024)

Education

George Mason University Fairfax, VA
Master of Science in Data Analytics Aug 2024 – May 2026

- Coursework: Machine Learning, Artificial Intelligence, Data Warehousing

VNR VJIET Hyderabad, India
Bachelor of Science in Computer Science Aug 2020 – May 2024

- Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management

Publications and Patents

- Restricting Unauthorized Access Using AWS Cloud
- Knowledge Discovery Approach Using Knowledge Engineering & Ontology
- PLANET-APP — Recommender System to treat individuals suffering from deficiency of Cognitive Intelligence Application No: 202541057363A— Indian Patent Published